

Cor Borde Bola Pseudohiperbolica

Desigualdades Auxiliares

Corolario 1. Sean $z_1, z_2 \in \mathbb{D}$, entonces

$$(1) \frac{z_1 - z_2}{1 - \overline{z_1} z_2} \in \partial B_\varrho(z_1, |z_2|) \quad (2) \frac{|z_1| - |z_2|}{1 - |z_1| |z_2|} \leq \left| \frac{z_1 - z_2}{1 - \overline{z_1} z_2} \right| \leq \frac{|z_1| + |z_2|}{1 + |z_1| |z_2|}.$$

Demostración:

(1) Basta ver que $\varrho(\tau_{z_1}(z_2), z_1) = |z_2|$. Como $\tau_{z_1} \in \text{Aut}(\mathbb{D})$, por el teorema de Schwarz-Pick, τ_{z_1} es una isometría para la métrica pseudohiperbólica. Por tanto, $\varrho(\tau_{z_1}(z_2), z_1) = \varrho(\tau_{z_1}(z_2), \tau_{z_1}(0)) = \varrho(z_2, 0) = |z_2|$.

(2) Se obtiene aplicando el `lem-cota-modulo-bola-pseudohiperbolica`/Lema 1 con $z = z_1$ y $r = |z_2|$ y usando ??.

■

Referenciado en

- Teo Metrica Pseudohiperbolica Desigualdad Triangular Generalizada

Temporary page!

L^AT_EX was unable to guess the total number of pages correctly. As there was some unprocessed data that should have been added to the final page this extra page has been added to receive it.

If you rerun the document (without altering it) this surplus page will go away, because L^AT_EX now knows how many pages to expect for this document.